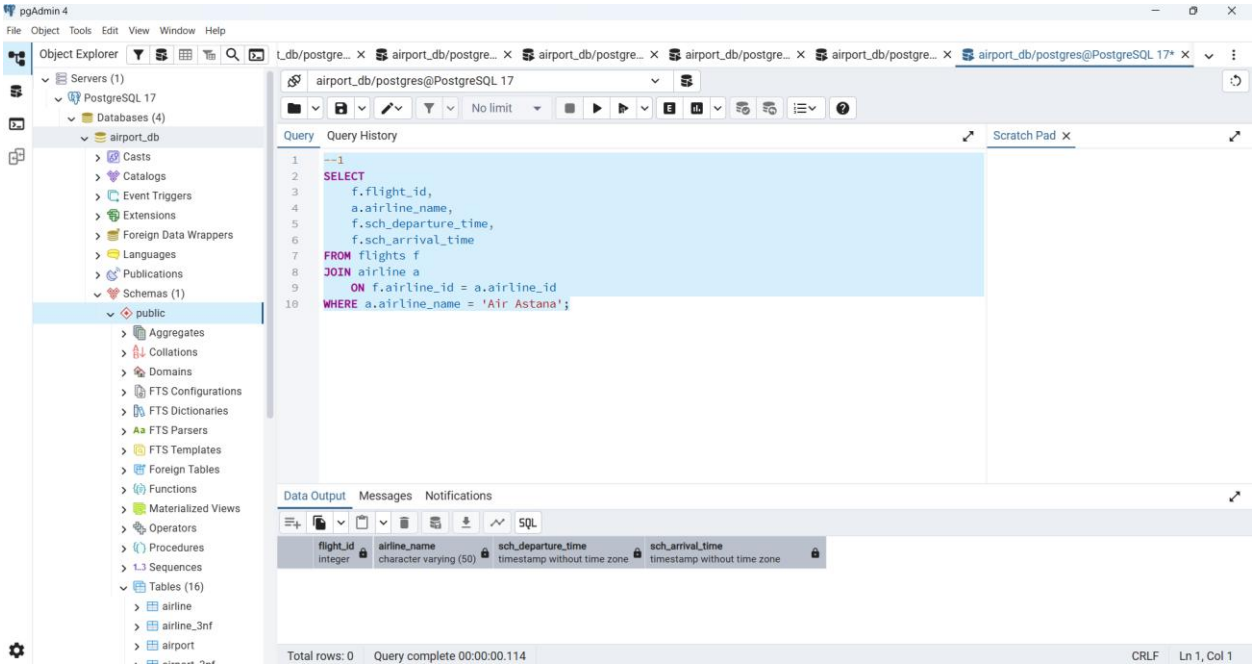
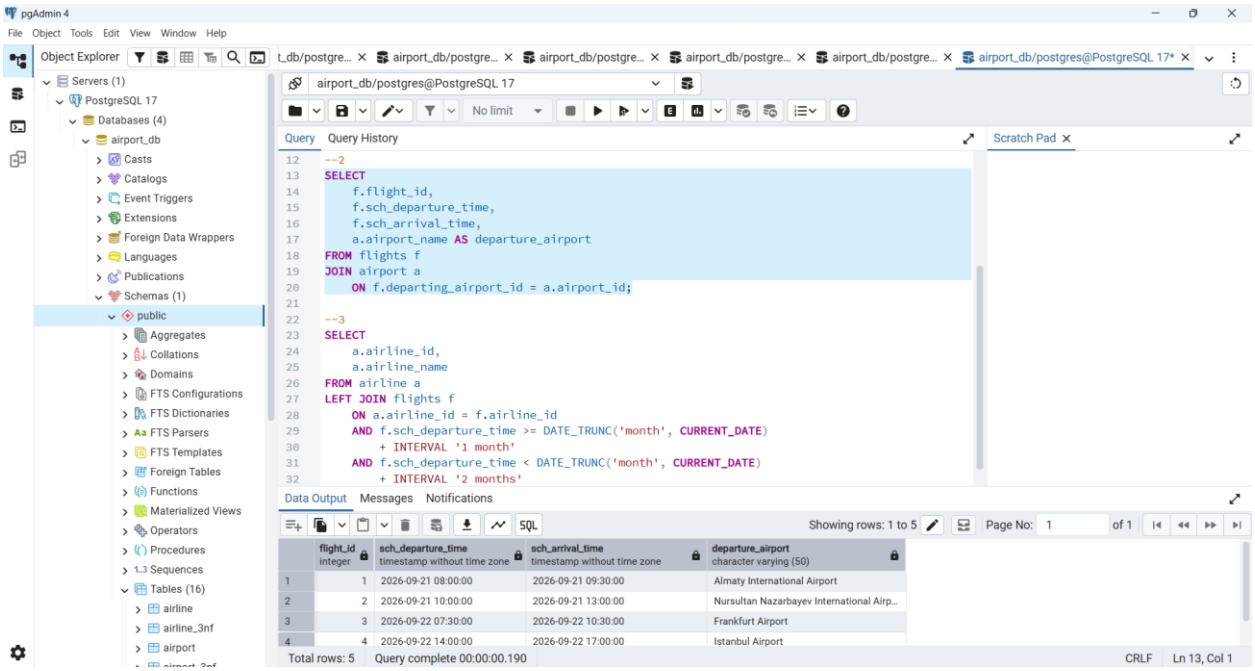
Laboratory work 6

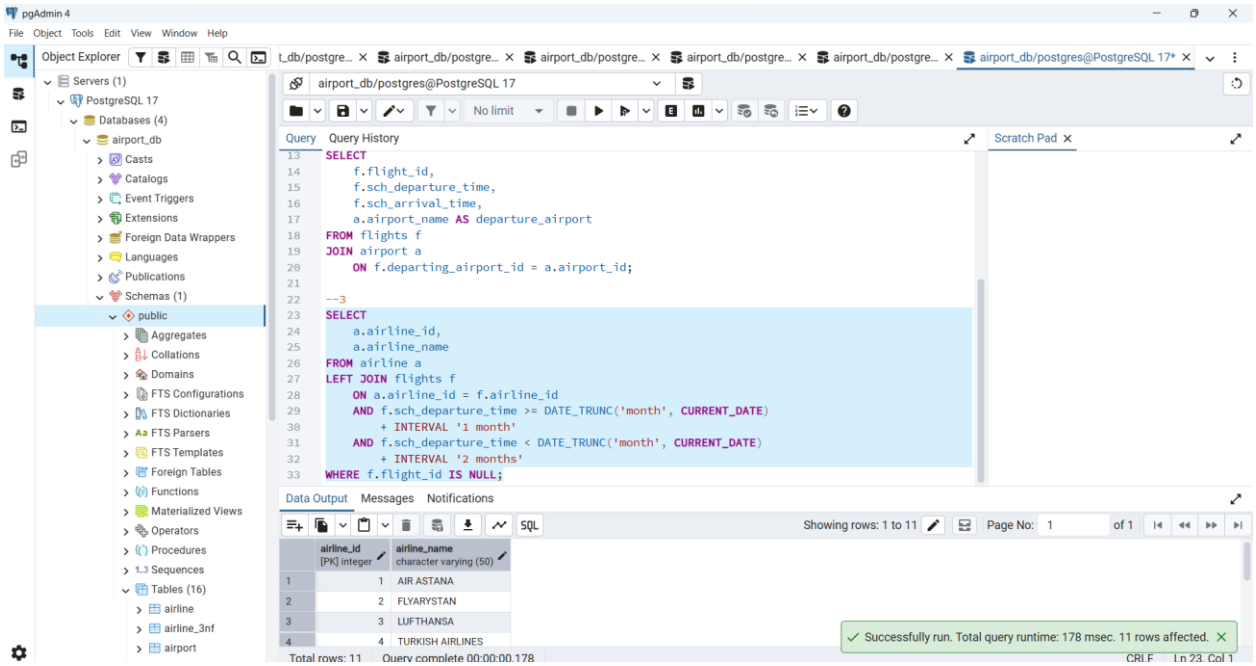
1



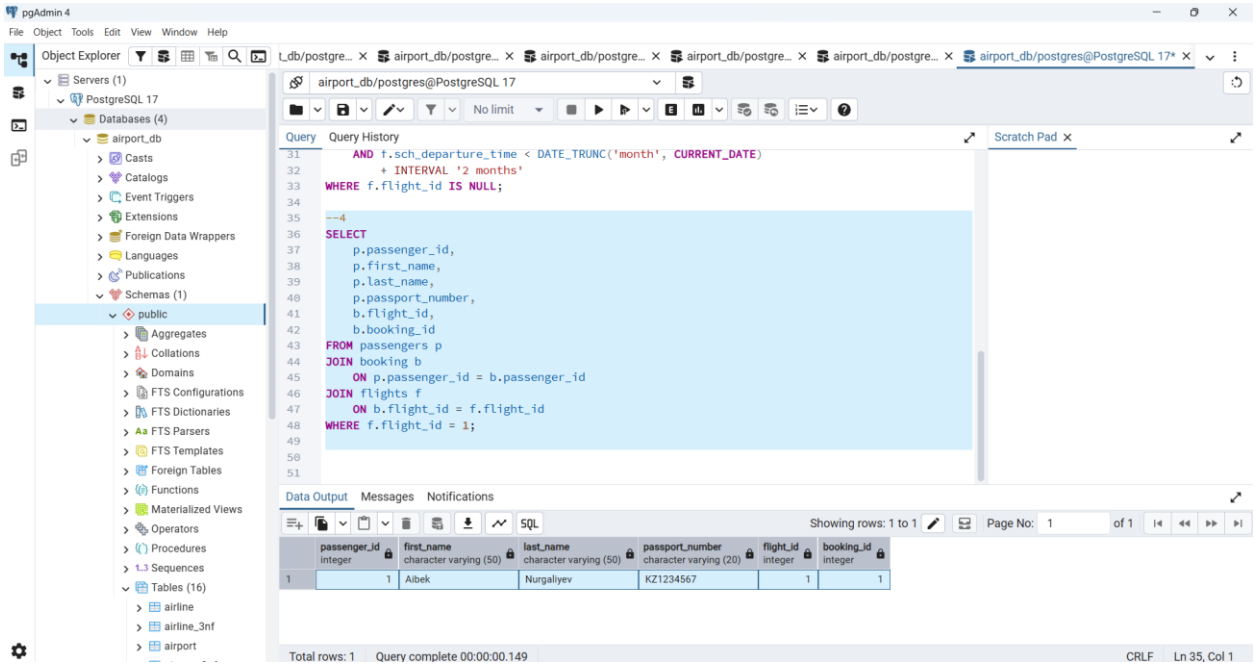
2



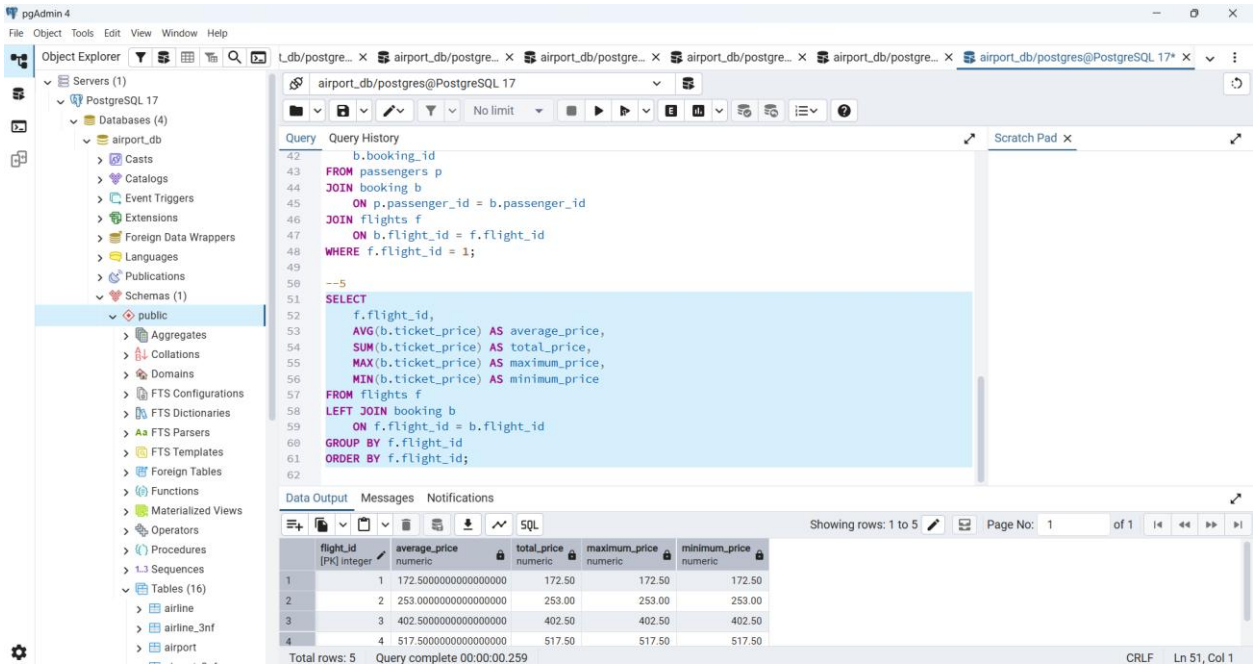
3


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
SELECT
f.flight_id,
f.sch_departure_time,
f.sch_arrival_time,
a.airport_name AS departure_airport
FROM flights f
JOIN airport a
ON f.departing_airport_id = a.airport_id;
--3
SELECT
a.airline_id,
a.airline_name
FROM airline a
LEFT JOIN flights f
ON a.airline_id = f.airline_id
AND f.sch_departure_time >= DATE_TRUNC('month', CURRENT_DATE)
+ INTERVAL '1 month'
AND f.sch_departure_time < DATE_TRUNC('month', CURRENT_DATE)
+ INTERVAL '2 months'
WHERE f.flight_id IS NULL;
Data Output Messages Notifications
Showing rows: 1 to 11
Page No: 1 of 1
airline_id [PK] integer
airline_name character varying (50)
1 1 AIR ASTANA
2 2 FLYARYSTAN
3 3 LUFTHANSA
4 4 TURKISH AIRLINES
Successfully run. Total query runtime: 178 msec. 11 rows affected.
Total rows: 11
Query complete 00:00:00.178
CRLF
Ln 23, Col 1

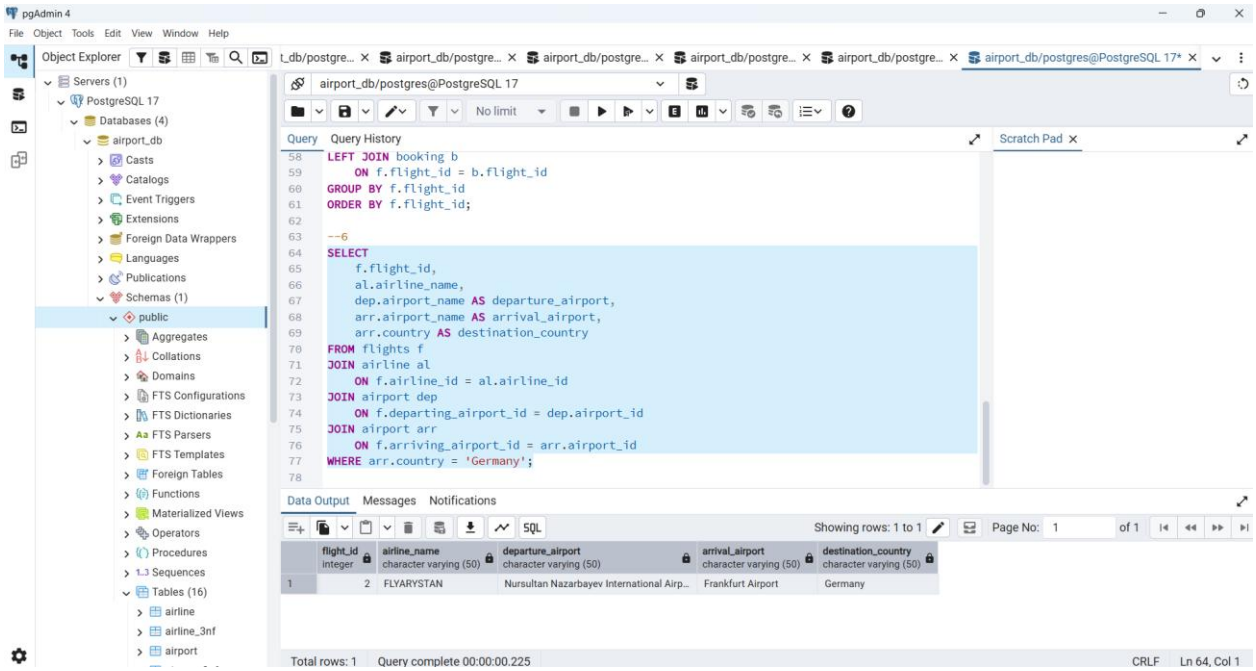
4


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
AND f.sch_departure_time < DATE_TRUNC('month', CURRENT_DATE)
+ INTERVAL '2 months'
WHERE f.flight_id IS NULL;
--4
SELECT
p.passenger_id,
p.first_name,
p.last_name,
p.passport_number,
b.flight_id,
b.booking_id
FROM passengers p
JOIN booking b
ON p.passenger_id = b.passenger_id
JOIN flights f
ON b.flight_id = f.flight_id
WHERE f.flight_id = 1;
Data Output Messages Notifications
Showing rows: 1 to 1
Page No: 1 of 1
passenger_id integer
first_name character varying (50)
last_name character varying (50)
passport_number character varying (20)
flight_id integer
booking_id integer
1 1 Aibek Nurgaliyev KZ1234567 1 1
Total rows: 1
Query complete 00:00:00.149
CRLF
Ln 35, Col 1

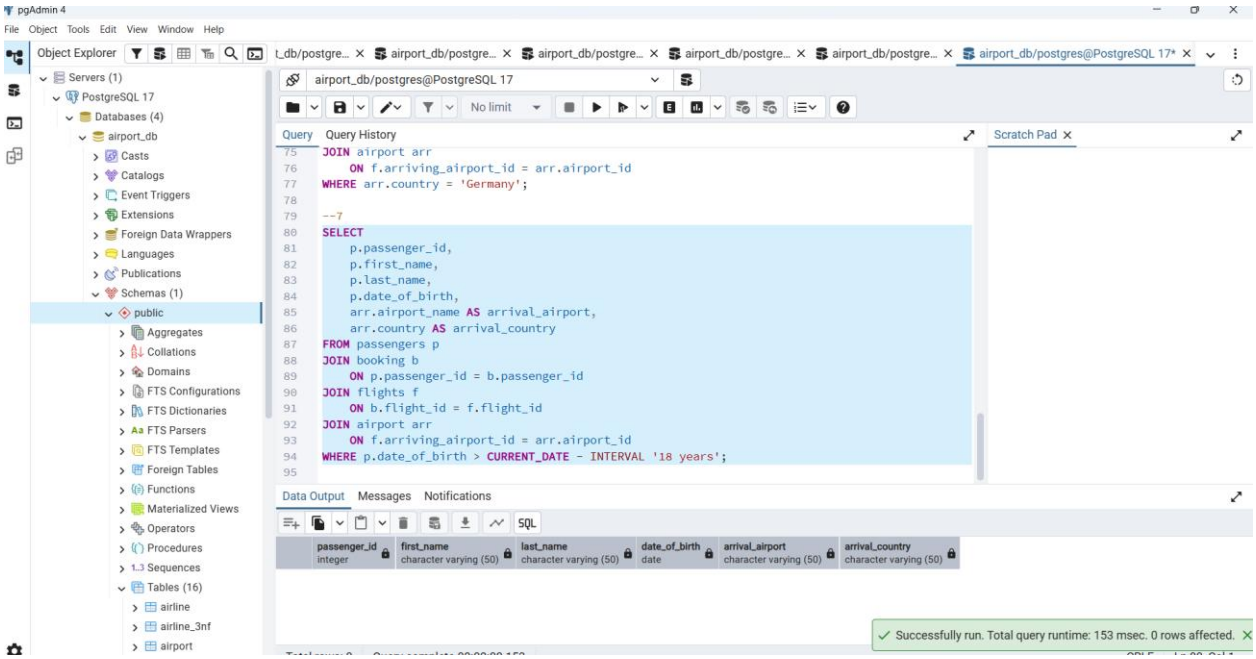
5


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
No limit
b.booking_id
FROM passengers p
JOIN booking b
ON p.passenger_id = b.passenger_id
JOIN flights f
ON b.flight_id = f.flight_id
WHERE f.flight_id = 1;
--5
SELECT
f.flight_id,
AVG(b.ticket_price) AS average_price,
SUM(b.ticket_price) AS total_price,
MAX(b.ticket_price) AS maximum_price,
MIN(b.ticket_price) AS minimum_price
FROM flights f
LEFT JOIN booking b
ON f.flight_id = b.flight_id
GROUP BY f.flight_id
ORDER BY f.flight_id;
Data Output Messages Notifications
Showing rows: 1 to 5
Page No: 1 of 1
flight_id [PK] integer
average_price numeric
total_price numeric
maximum_price numeric
minimum_price numeric
1 1 172.5000000000000000 172.50 172.50 172.50
2 2 253.0000000000000000 253.00 253.00 253.00
3 3 402.5000000000000000 402.50 402.50 402.50
4 4 517.5000000000000000 517.50 517.50 517.50
Total rows: 5
Query complete 00:00:00.259
CRLF
Ln 51, Col 1

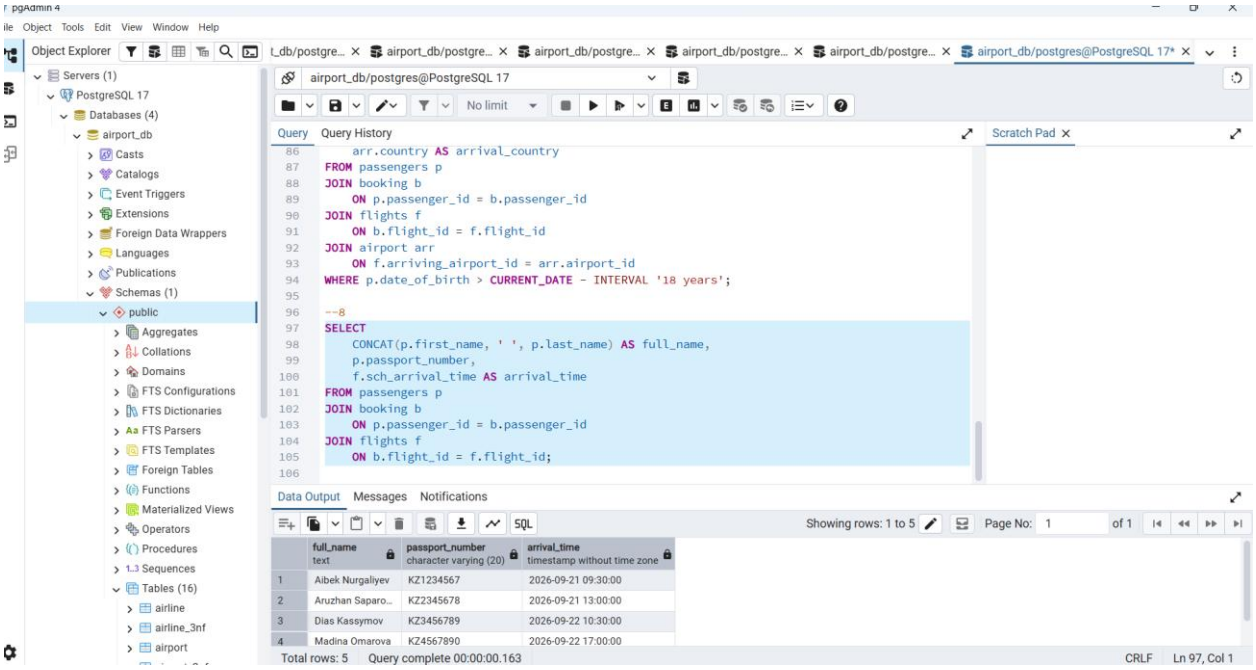
6


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
No limit
LEFT JOIN booking b
ON f.flight_id = b.flight_id
GROUP BY f.flight_id
ORDER BY f.flight_id;
--6
SELECT
f.flight_id,
al.airline_name,
dep.airport_name AS departure_airport,
arr.airport_name AS arrival_airport,
arr.country AS destination_country
FROM flights f
JOIN airline al
ON f.airline_id = al.airline_id
JOIN airport dep
ON f.departing_airport_id = dep.airport_id
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
WHERE arr.country = 'Germany';
Data Output Messages Notifications
Showing rows: 1 to 1
Page No: 1 of 1
flight_id integer
airline_name character varying (50)
departure_airport character varying (50)
arrival_airport character varying (50)
destination_country character varying (50)
1 2 FLYARYSTAN Nursultan Nazarbayev International Airp... Frankfurt Airport Germany
Total rows: 1
Query complete 00:00:00.225
CRLF
Ln 64, Col 1

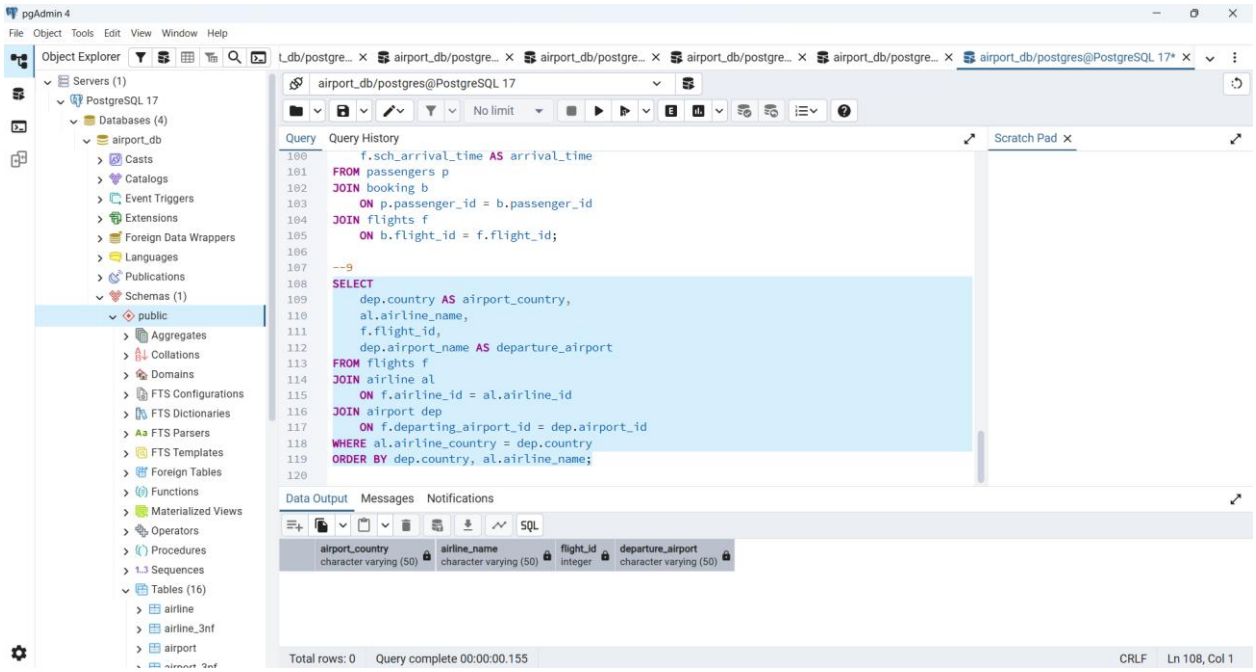
7


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17*
Servers (1)
PostgreSQL 17
Databases (4)
airport_db
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Domains
FTS Configurations
FTS Dictionaries
FTS Parsers
FTS Templates
Foreign Tables
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Procedures
Sequences
Tables (16)
airline
airline_3nf
airport
airport_db/postgres@PostgreSQL 17
No limit
Query
Query History
Scratch Pad
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
WHERE arr.country = 'Germany';
--7
SELECT
p.passenger_id,
p.first_name,
p.last_name,
p.date_of_birth,
arr.airport_name AS arrival_airport,
arr.country AS arrival_country
FROM passengers p
JOIN booking b
ON p.passenger_id = b.passenger_id
JOIN flights f
ON b.flight_id = f.flight_id
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
WHERE p.date_of_birth > CURRENT_DATE - INTERVAL '18 years';
Data Output
Messages
Notifications
SQL
passenger_id integer
first_name character varying (50)
last_name character varying (50)
date_of_birth date
arrival_airport character varying (50)
arrival_country character varying (50)
Successfully run. Total query runtime: 153 msec. 0 rows affected.

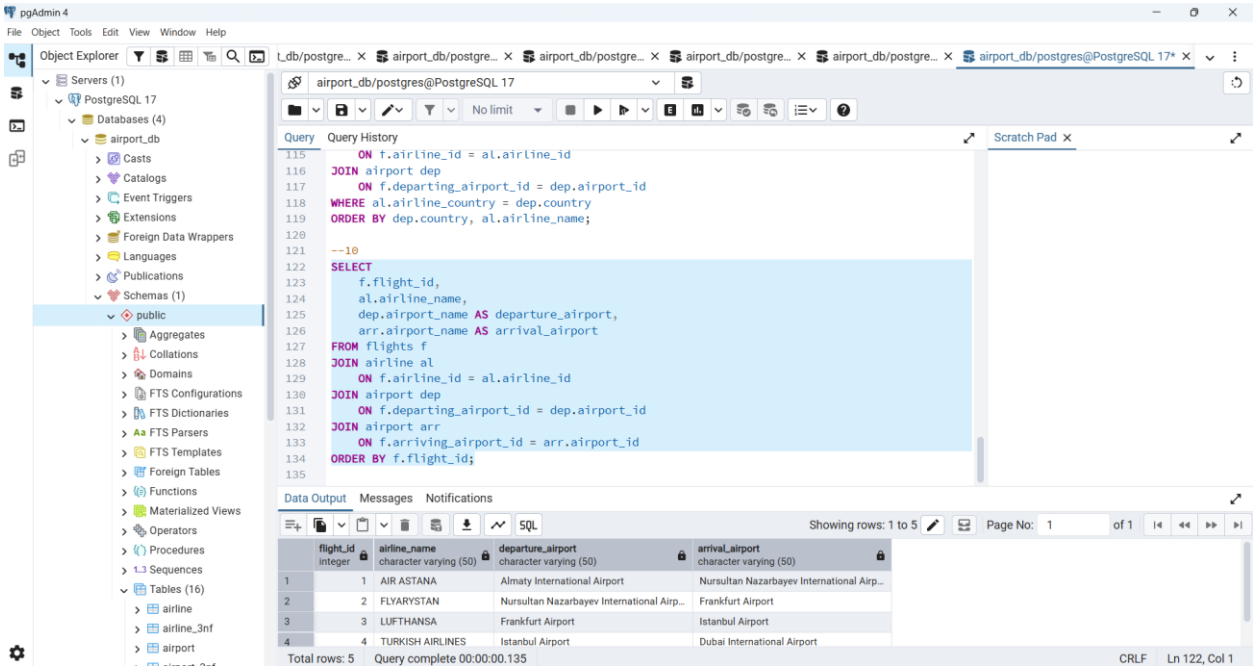
8


Object Explorer
airport_db/postgres@PostgreSQL 17*
airport_db/postgres@PostgreSQL 17
No limit
Query
Query History
Scratch Pad
arr.country AS arrival_country
FROM passengers p
JOIN booking b
ON p.passenger_id = b.passenger_id
JOIN flights f
ON b.flight_id = f.flight_id
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
WHERE p.date_of_birth > CURRENT_DATE - INTERVAL '18 years';
--8
SELECT
CONCAT(p.first_name, ' ', p.last_name) AS full_name,
p.passport_number,
f.sch_arrival_time AS arrival_time
FROM passengers p
JOIN booking b
ON p.passenger_id = b.passenger_id
JOIN flights f
ON b.flight_id = f.flight_id;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 5
Page No: 1
of 1
full_name text
passport_number character varying (20)
arrival_time timestamp without time zone
1 Aibek Nurgaliyev KZ1234567 2026-09-21 09:30:00
2 Aruzhan Saparo... KZ2345678 2026-09-21 13:00:00
3 Dias Kassymov KZ3456789 2026-09-22 10:30:00
4 Madina Omarova KZ4567890 2026-09-22 17:00:00
Total rows: 5
Query complete 00:00:00.163
CRLF
Ln 97, Col 1

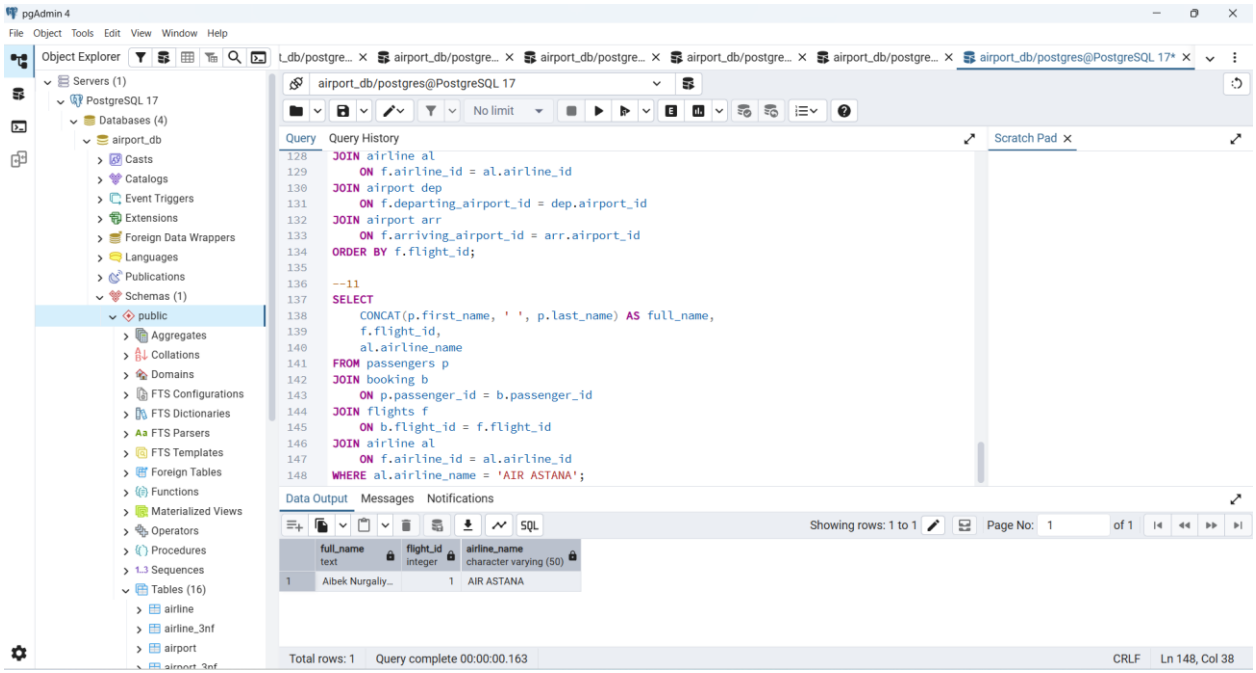
9


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
f.sch_arrival_time AS arrival_time
FROM passengers p
JOIN booking b
ON p.passenger_id = b.passenger_id
JOIN flights f
ON b.flight_id = f.flight_id;
--9
SELECT
dep.country AS airport_country,
al.airline_name,
f.flight_id,
dep.airport_name AS departure_airport
FROM flights f
JOIN airline al
ON f.airline_id = al.airline_id
JOIN airport dep
ON f.departing_airport_id = dep.airport_id
WHERE al.airline_country = dep.country
ORDER BY dep.country, al.airline_name;
Data Output Messages Notifications
airport_country character varying (50)
airline_name character varying (50)
flight_id integer
departure_airport character varying (50)
Total rows: 0 Query complete 00:00:00.155
CRLF Ln 108, Col 1

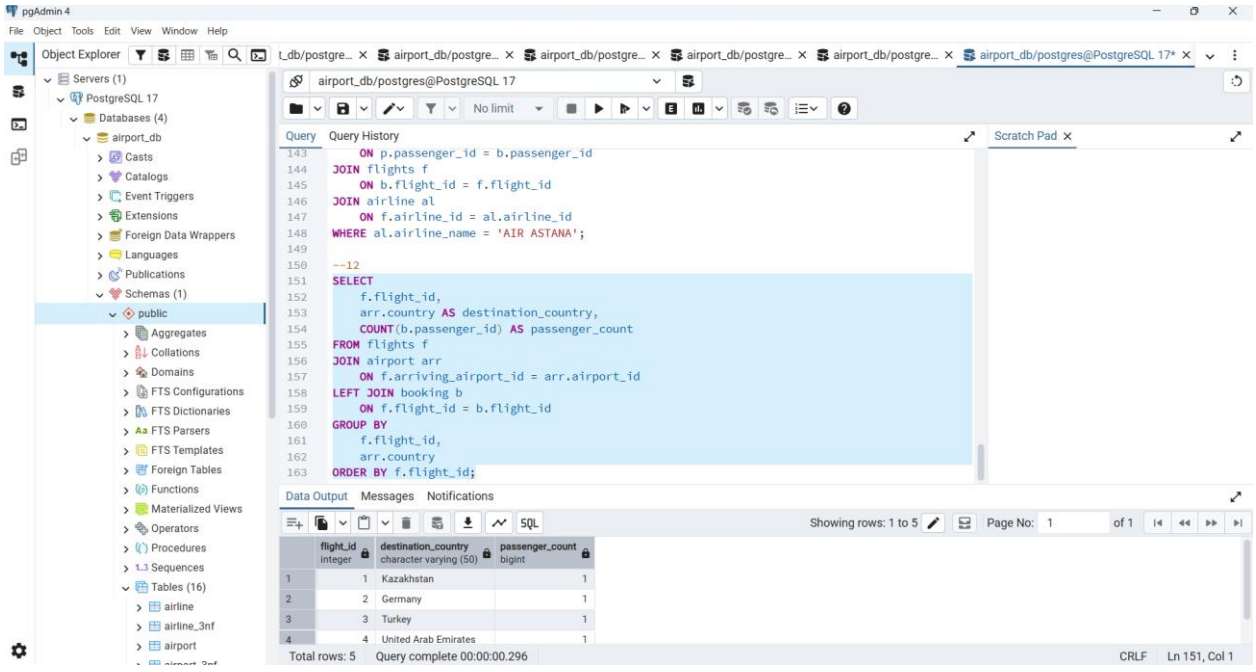
10


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
ON f.airline_id = al.airline_id
JOIN airport dep
ON f.departing_airport_id = dep.airport_id
WHERE al.airline_country = dep.country
ORDER BY dep.country, al.airline_name;
--10
SELECT
f.flight_id,
al.airline_name,
dep.airport_name AS departure_airport,
arr.airport_name AS arrival_airport
FROM flights f
JOIN airline al
ON f.airline_id = al.airline_id
JOIN airport dep
ON f.departing_airport_id = dep.airport_id
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
ORDER BY f.flight_id;
Data Output Messages Notifications
Showing rows: 1 to 5
Page No: 1 of 1
flight_id integer
airline_name character varying (50)
departure_airport character varying (50)
arrival_airport character varying (50)
1 1 AIR ASTANA Almaty International Airport Nursultan Nazarbayev International Airp...
2 2 FLYARYSTAN Nursultan Nazarbayev International Airp... Frankfurt Airport
3 3 LUFTHANSA Frankfurt Airport Istanbul Airport
4 4 TURKISH AIRLINES Istanbul Airport Dubai International Airport
Total rows: 5 Query complete 00:00:00.135
CRLF Ln 122, Col 1

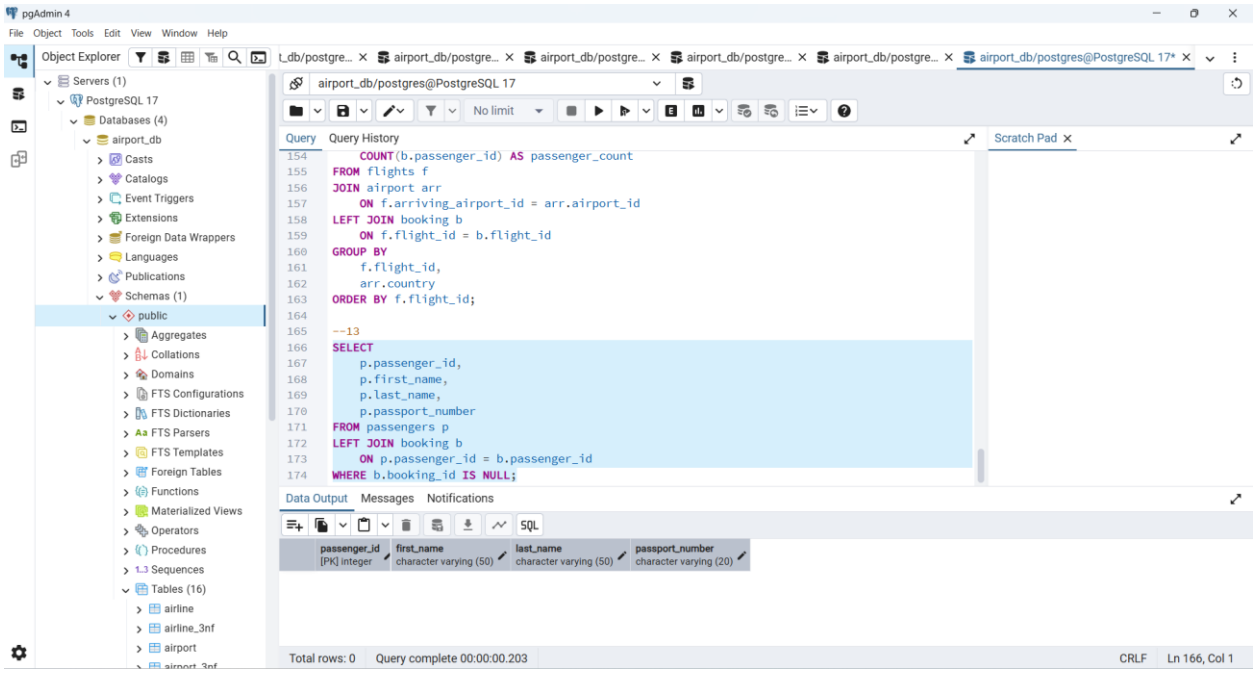
11


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
No limit
--11
SELECT
CONCAT(p.first_name, ' ', p.last_name) AS full_name,
f.flight_id,
al.airline_name
FROM passengers p
JOIN booking b
ON p.passenger_id = b.passenger_id
JOIN flights f
ON b.flight_id = f.flight_id
JOIN airline al
ON f.airline_id = al.airline_id
WHERE al.airline_name = 'AIR ASTANA';
Data Output Messages Notifications
Showing rows: 1 to 1
Page No: 1 of 1
full_name text
flight_id integer
airline_name character varying (50)
1 Aibek Nurgaliy... 1 AIR ASTANA
Total rows: 1 Query complete 00:00:00.163
CRLF Ln 148, Col 38

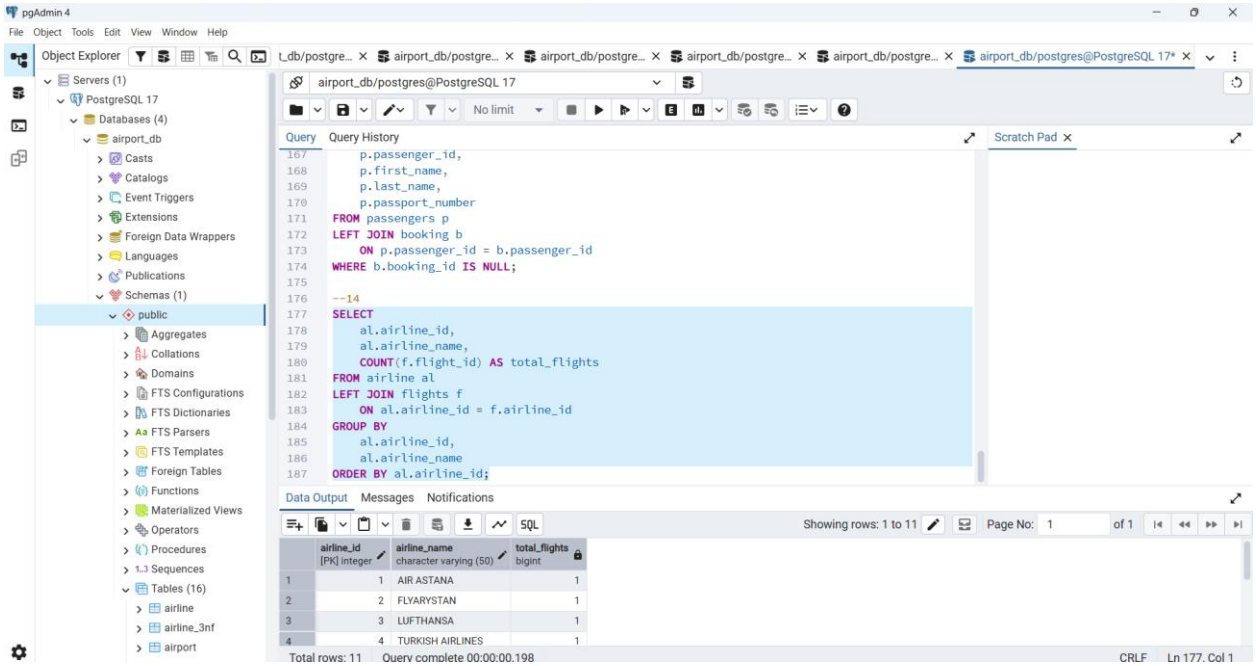
12


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Query Query History
Scratch Pad
No limit
--12
SELECT
f.flight_id,
arr.country AS destination_country,
COUNT(b.passenger_id) AS passenger_count
FROM flights f
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
LEFT JOIN booking b
ON f.flight_id = b.flight_id
GROUP BY
f.flight_id,
arr.country
ORDER BY f.flight_id;
Data Output Messages Notifications
Showing rows: 1 to 5
Page No: 1 of 1
flight_id integer
destination_country character varying (50)
passenger_count bigint
1 1 Kazakhstan 1
2 2 Germany 1
3 3 Turkey 1
4 4 United Arab Emirates 1
Total rows: 5 Query complete 00:00:00.296
CRLF Ln 151, Col 1

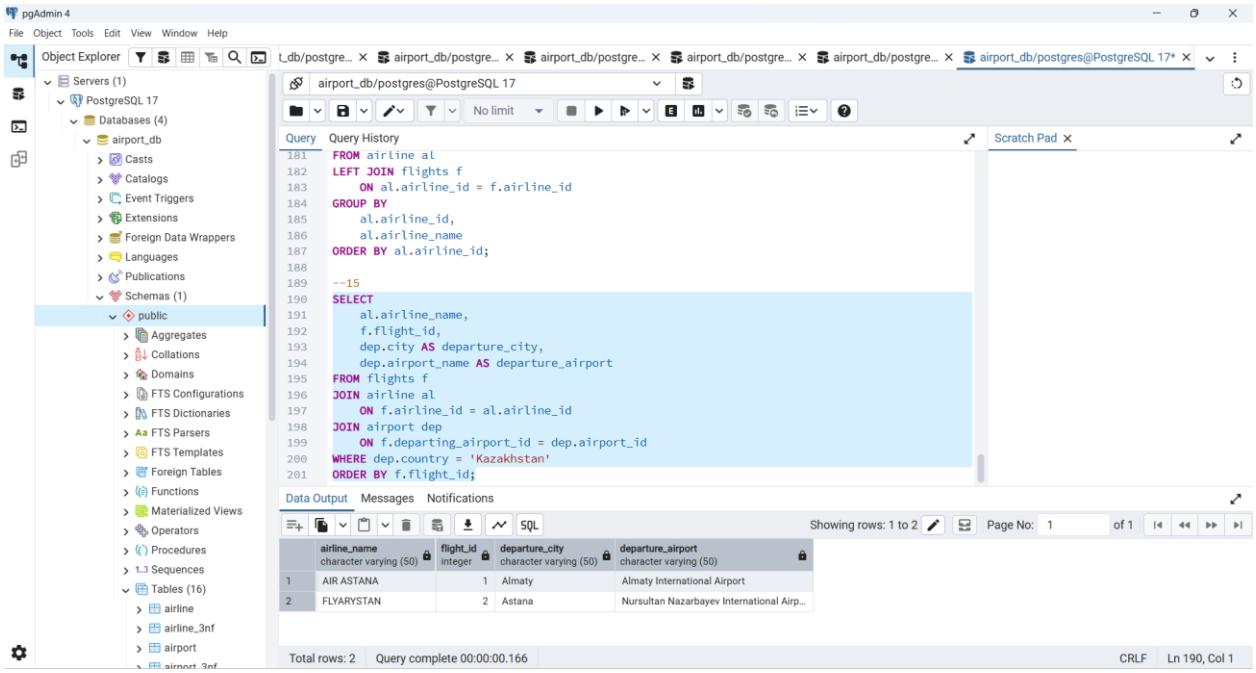
13


pgAdmin 4
File Object Tools Edit View Window Help
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airport_db/postgres@PostgreSQL 17*
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PostgreSQL 17
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FTS Configurations
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Sequences
Tables (16)
airline
airline_3nf
airport
Query
Query History
Scratch Pad
COUNT(b.passenger_id) AS passenger_count
FROM flights f
JOIN airport arr
ON f.arriving_airport_id = arr.airport_id
LEFT JOIN booking b
ON f.flight_id = b.flight_id
GROUP BY
f.flight_id,
arr.country
ORDER BY f.flight_id;
--13
SELECT
p.passenger_id,
p.first_name,
p.last_name,
p.passport_number
FROM passengers p
LEFT JOIN booking b
ON p.passenger_id = b.passenger_id
WHERE b.booking_id IS NULL;
Data Output
Messages
Notifications
passenger_id [PK] integer
first_name character varying (50)
last_name character varying (50)
passport_number character varying (20)
Total rows: 0
Query complete 00:00:00.203
CRLF
Ln 166, Col 1

14


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17*
Servers (1)
PostgreSQL 17
Databases (4)
airport_db
Casts
Catalogs
Event Triggers
Extensions
Foreign Data Wrappers
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Schemas (1)
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FTS Parsers
FTS Templates
Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (16)
airline
airline_3nf
airport
Query
Query History
Scratch Pad
p.passenger_id,
p.first_name,
p.last_name,
p.passport_number
FROM passengers p
LEFT JOIN booking b
ON p.passenger_id = b.passenger_id
WHERE b.booking_id IS NULL;
--14
SELECT
al.airline_id,
al.airline_name,
COUNT(f.flight_id) AS total_flights
FROM airline al
LEFT JOIN flights f
ON al.airline_id = f.airline_id
GROUP BY
al.airline_id,
al.airline_name
ORDER BY al.airline_id;
Data Output
Messages
Notifications
Showing rows: 1 to 11
Page No: 1
of 1
airline_id [PK] integer
airline_name character varying (50)
total_flights bigint
1 1 AIR ASTANA 1
2 2 FLYARYSTAN 1
3 3 LUFTHANSA 1
4 4 TURKISH AIRLINES 1
Total rows: 11
Query complete 00:00:00.198
CRLF
Ln 177, Col 1

15


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport_db/postgres@PostgreSQL 17
Servers (1)
PostgreSQL 17
Databases (4)
airport_db
Casts
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Event Triggers
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Foreign Tables
Functions
Materialized Views
Operators
Procedures
Sequences
Tables (16)
airline
airline_3nf
airport
Query
Query History
Scratch Pad
FROM airline al
LEFT JOIN flights f
ON al.airline_id = f.airline_id
GROUP BY
al.airline_id,
al.airline_name
ORDER BY al.airline_id;
--15
SELECT
al.airline_name,
f.flight_id,
dep.city AS departure_city,
dep.airport_name AS departure_airport
FROM flights f
JOIN airline al
ON f.airline_id = al.airline_id
JOIN airport dep
ON f.departing_airport_id = dep.airport_id
WHERE dep.country = 'Kazakhstan'
ORDER BY f.flight_id;
Data Output
Messages
Notifications
Showing rows: 1 to 2
Page No: 1
of 1
airline_name character varying (50)
flight_id integer
departure_city character varying (50)
departure_airport character varying (50)
1 AIR ASTANA 1 Almaty Almaty International Airport
2 FLYARYSTAN 2 Astana Nursultan Nazarbayev International Airp...
Total rows: 2
Query complete 00:00:00.166
CRLF
Ln 190, Col 1